Conics: Transformations

Questions 1–23

Question 1. Write down the equation of the image of the curve $y^2 = 4x$ when it has been transformed by each of the following transformations:

a) A stretch, parallel to the x-axis, scale factor 3
b) A stretch, parallel to the y-axis, scale factor $\frac{1}{4}$
c) A translation by the vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$
d) A reflection in the x-axis.
e) A reflection in the line $y = x$

Question 2. Write down the equation of the image of the curve $x^2 + 3y^2 = 3$ when it has been transformed by each of the following transformations:

a) A stretch, parallel to the x-axis, scale factor $\frac{1}{2}$
b) A stretch, parallel to the y-axis, scale factor 5
c) A translation by the vector $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$
d) A reflection in the y-axis.
e) A reflection in the line $y = -x$

Question 3. Write down the equation of the image of the curve $xy = 12$ when it has been transformed by each of the following transformations:

a) A stretch, parallel to the x-axis, scale factor 3
b) A stretch, parallel to the y-axis, scale factor 2
c) A translation by the vector $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$
d) A reflection in the x-axis.
e) A reflection in the line $y = -x$

Question 4. Write down the equation of the image of the curve given by $\frac{x^2}{4} - \frac{y^2}{5} = 1$ when it has been transformed by each of the following transformations:

a) A stretch, parallel to the x-axis, scale factor $\frac{1}{4}$
b) A stretch, parallel to the y-axis, scale factor 2
c) A translation by the vector $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$
d) A reflection in the x-axis.
e) A reflection in the line $y = x$

Question 5. Each of the following curves are transformed by a stretch, parallel to the x-axis scale factor 2 Find the equation of the image and state the name of its shape:

a) $\frac{x^2}{3} + \frac{y^2}{16} = 1$
b) $\frac{x^2}{2} - \frac{y^2}{10} = 1$
c) $xy = 14$
d) $x^2 - y^2 = 4$

Question 6. Each of the following curves are transformed by a stretch, parallel to the y-axis scale factor 3 Find the equation of the image and state the name of its shape:

a) $\frac{x^2}{18} + \frac{y^2}{2} = 1$
b) $\frac{x^2}{45} - \frac{y^2}{5} = 1$
c) $y^2 = -3x$
d) $x^2 - y^2 = 2$

Question 7. A circle with equation $x^2+y^2=4$ is transformed by the given transformation. In each case, determine the equation of the ellipse it forms in the form $\frac{x^2}{a}+\frac{y^2}{b}=1$ where a and b are constants to be found.

a) A stretch, parallel to the x-axis, scale factor 2

b) A stretch, parallel to the y-axis, scale factor 2

Question 8. Each of the following curves are reflected in the line $y=x$. In each case, find the equation of the image of the curve:

a) $y^2=\frac{1}{2}x$

b) $3x^2+7y^2=21$

c) $xy=-6$

d) $\frac{x^2}{8}-\frac{y^2}{4}=1$

Question 9. Each of the following curves are reflected in the line $y = -x$. In each case, find the equation of the image of the curve:

a) $y^2 = \frac{1}{4}x$

b) $2x^2 + 5y^2 = 10$

c) $xy = \frac{1}{5}$

d) $\frac{x^2}{3} - \frac{y^2}{25} = 1$

Prerequisite knowledge for Q10-Q11: Matrix Transformations

Question 10. The curve C_1 has equation $\frac{x^2}{4} + \frac{y^2}{8} = 1$

The matrix $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ maps C_1 onto C_2

a) Find the equation of the curve C_2

The matrix $\mathbf{N} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ maps C_1 onto C_3

b) Find the equation of the curve C_3

Question 11. The curve C_1 has equation $\frac{x^2}{12} - \frac{y^2}{4} = 1$

The matrix $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ maps C_1 onto C_2

a) Find the equation of the curve C_2

b) Find the asymptotes of curve C_2

The matrix $\mathbf{N} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ maps C_1 onto C_3

c) Find the equation of the curve C_3

d) Find the asymptotes of curve C_3

Question 12. Describe the transformation which maps:

a) $y^2 = 2x$ to $(y-3)^2 = 2x$

b) $\frac{x^2}{4} - \frac{y^2}{3} = 1$ to $\frac{x^2}{4} - \frac{(y+1)^2}{3} = 1$

c) $\frac{x^2}{9} + \frac{y^2}{16} = 1$ to $\frac{(x-1)^2}{9} + \frac{(y+3)^2}{16} = 1$

d) $\frac{x^2}{16} + \frac{y^2}{2} = 1$ to $\frac{x^2}{4} + \frac{y^2}{2} = 1$

e) $\frac{x^2}{2} - \frac{y^2}{5} = 1$ to $\frac{x^2}{18} - \frac{y^2}{5} = 1$

f) $\frac{x^2}{16} - y^2 = 4$ to $x^2 - 16(y+2)^2 = 64$

g) $\frac{x^2}{8} - \frac{y^2}{2} = 1$ to $(x+5)^2 - 4(y-1)^2 = 8$

h) $\frac{x^2}{16} - \frac{y^2}{2} = 1$ to $x^2 - 288y^2 = 16$

i) $\frac{x^2}{6} - \frac{y^2}{4} = 1$ to $2x^2 - 12x - 3y^2 - 18y = 21$

Question 13. Sketch each of the following curves, label each intersection with the axes clearly:

a) $y^2 = 4(x+1)$
b) $(y-4)^2 = 2x$
c) $(y-3)^2 = 8(x+2)$
d) $y^2 = 2x - 6$

Question 14. Sketch each of the following curves, label each intersection with the axes clearly and include any asymptotes in your sketch.

a) $x(y-2) = 6$
b) $y(x+5) = -15$
c) $(x-2)(y+4) = 12$
d) $3y + xy = 9$
e) $xy - 2y = 5$

Question 15. The curve C_1 given by equation $\frac{x^2}{25} + \frac{y^2}{5} = 1$ is mapped to curve C_2 with equation

$$x^2 + 5(y+1)^2 = 25$$

a) Describe the transformation which maps curve C_1 to curve C_2
b) Find where the curve C_2 intersects with the coordinate axes.
c) Hence, sketch the curve C_2

Question 16. A curve C is given by equation $\frac{x^2}{9} - \frac{y^2}{12} = 1$

a) State the type of curve C is.
b) State the asymptotes of C
c) Given that C is translated by $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$, find the asymptotes of the image of C

Question 17. The curve C_1 has equation $\frac{x^2}{4} - \frac{y^2}{25} = 1$

a) Sketch the curve C_1

The curve C_1 is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$ where k is a positive constant to give curve C_2

The curve C_2 passes through the origin.

b) State the value of k

c) Find the equations of the asymptotes of C_2 giving your answer in the form $y = mx + c$

Question 18. The curve C_1 given by equation $\frac{x^2}{9} + y^2 = 1$ is mapped to curve C_2 with equation

$$x^2 - 4x + 9y^2 = 5$$

a) Find where the curve C_1 intersects with the coordinate axes.

b) Describe the transformation which maps curve C_1 to curve C_2

c) Find the coordinates of the maximum and minimum values of x and y on the curve C_2

d) Find where the curve C_2 intersects with the coordinate axes.

e) Hence, sketch the curve C_2

Question 19. The curve C_1 given by equation $\frac{x^2}{9} - \frac{y^2}{4} = 1$ is mapped to curve C_2 with equation

$$\frac{(x+4)^2}{9} - \frac{y^2}{4} = 1$$

a) State the equations of the asymptotes of curve C_1
b) Describe the transformation which maps curve C_1 to curve C_2
c) State the equations of the asymptotes of curve C_2
d) Find where the curve C_2 intersects with the coordinate axes.
e) Hence, sketch the curve C_2 labelling where the asymptotes intersect the axes.

Question 20. The curve C_1 given by equation $\frac{x^2}{4} - \frac{(y-2)^2}{25} = 1$ is mapped to curve C_2 with equation

$$\frac{x^2}{4} - \frac{(y+3)^2}{25} = 1$$

a) Determine the equations of the asymptotes of curve C_1
b) Describe the transformation which maps curve C_1 to curve C_2
c) Find the asymptotes of the curve C_2
d) Find where the curve C_2 intersects with the coordinate axes.
e) Hence, sketch the curve C_2 labelling where the asymptotes intersect the axes.

Question 21. The curve C_1 is given by the equation

$$\frac{x^2}{9} + \frac{y^2}{25} = 1$$

The curve C_2 is the image of C_1 after a translation, such that C_2 touches the x-axis at a single positive value of x and it touches the y-axis at a single positive value of y

a) Find the vector that translates C_1 to C_2
b) State the points of intersection of C_1 and C_2
c) Sketch both C_1 and C_2 on the same axes.
d) Find the gradient m of the equations of the tangent lines which are a tangent line to both C_1 and C_2
e) Show that for your m the tangent lines are $y = mx + 5\sqrt{2}$ and $y = mx - 5\sqrt{2}$

Question 22. An ellipse E is given by the equation

$$\frac{x^2}{8} + \frac{y^2}{3} = 1$$

A straight line L is given by the equation

$$y = 2x + 1$$

a) Write down the coordinates of the points where the ellipse E intersects the coordinate axes.

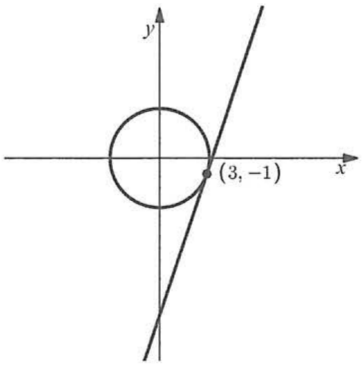
The ellipse E is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$ where k is a positive constant.

b) Write down the equation of the translated ellipse.
c) Show, that if the translated ellipse intersects the line L then the points of intersection must satisfy the equation

$$35x^2 + (32 - 6k)x + (3k^2 - 16) = 0$$

d) Given that the line L is a tangent to the translated ellipse, find the value of k

Question 23. The diagram shows a circle C and a line L, which is the tangent to C at the point $(3, -1)$



The equations of C and L_1 are

$$x^2 + y^2 = 10 \text{ and } 3x - y = 10$$

respectively.
The circle C is now transformed by a stretch with scale factor 3 parallel to the x-axis. The image of C under this stretch is an ellipse E. The line is also transformed by a stretch with scale factor 3 parallel to the x-axis to form line L_2

a) Find the equation of the ellipse E
b) Find the equation of the line L_2
c) Show that L_2 is a tangent to the ellipse E and hence find the point of intersection.
d) Sketch the ellipse E and the line L_2